TVS Automotive Solutions: Using data to deliver faster aftermarket services for drivers with Google Cloud

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Solutions

TVS Automobile Solutions

About TVS Automotive Solutions

Formed in 2011, TVS Automotive Solutions (TVS), a part of the TVS Group, is an automobile aftermarket service provider that offers quality parts and workmanship to help vehicle owners extend the value of their asset. TVS operates a network of garages and auto part suppliers to service cars, two-wheelers, and commercial vehicles across India. TVS connects garages with auto parts suppliers on its PartSmart online marketplace that carries parts and accessories from various leading brands. Consumers can

TVS migrates its premises ERP an workloads to Cor Engine for a digit that analyzes cor car data with Clo Bigtable, providin

to garages for fa
servicing.

order aftermarket services such as car repairs and insurance on myTVS app.

Industries: Automotive

Location: India

Google Cloud results

- Eliminates server-related downtime by moving on-premises business-critical applications to Compute Engine to scale cloud resources at any time
- Migrates more than 20 TB of transaction data to Cloud Storage for business continuity

33% re
mainte
time

Cloud IoT Core

About Niveus

Niveus has been a Google Cloud Premier Partner since 2019 and winner of the 2020 Breakthrough Partner of the Year - Asia Pacific award. It has a customer base that spans across the USA and India. Niveus operates in multiple cities, including Bengaluru, Delhi, Mumbai, and Singapore, with a team of engineers that specializes in building modern applications with Kubernetes containers in the cloud.

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Google Cloud (https://cloud.google.com/)

Cloud Bigtable (https://cloud.google.com/bigtable/)

Compute Engine (https://cloud.google.com/compute/)

Cloud IoT Core (https://cloud.google.com/iot-core/)

Cloud SQL (https://cloud.google.com/sql/)

Cloud Storage (https://cloud.google.com/storage/)

in case of disaster

- Simplifies IT operations with Cloud Logging and Cloud Monitoring to identify and resolve application problems quickly and efficiently
- Equips traditional IT staff with new skills on Google Cloud to support evolving business needs

Dataflow (https://cloud.google.com/dataflow/)

Pacific award. It has a customer base that

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

Cloud Pub/Sub (https://cloud.google.com/pubsub/)

Operations (https://cloud.google.com/products/operations)

Kubernetes containers in the cloud.

Logging (https://cloud.google.com/logging/)

Monitoring (https://cloud.google.com/monitoring/)

Poor servicing is often the top pain point for vehicle owners. From the time a car stalls to having it towed to a garage, to long waiting times for replacement parts and finding alternative means of transport, the entire experience can be quite stressful. With this in mind, TVS Automobile Solutions (TVS) (https://www.tvsautomobilesolutions.com/), an aftermarket service provider, aims to make servicing hassle free, so vehicle owners get first-time right service.

For over a decade, customers in India have relied on TVS for its aftermarket services, from a simple oil change to emergency roadside assistance for vehicles. Laying the groundwork for global expansion, TVS Automobile Solutions transforms its traditional aftermarket business to serve more customers faster, with a data-driven platform on Google Cloud (https://cloud.google.com/).

"Moving on-premises to cloud was a huge transition for us. Google Cloud stood out from other cloud providers with a well-articulated modernization plan and a network of experienced partners to help us get ROI very quickly," says Sai Satheesh, Chief Technology Officer at TVS. "Instead of pushing us to a set of technologies, Google Cloud spent time understanding our needs to deliver a solution that scales with our business."

TVS took four months to carry out the first two stages of its transformation journey. The first stage was to migrate enterprise resource planning (ERP) and its surrounding applications to Google Cloud. This involved working with a previous Google Cloud partner, who helped TVS migrate from on-premises to Google Cloud with minimal downtime and continue to provide ongoing managed services to optimize its cloud infrastructure.

The second stage was to modernize mobile applications for its parts store and auto repair service, before building a connected car platform. To achieve this, Niveus (https://niveussolutions.com/) helped TVS build its product catalog using Elasticsearch and Google Kubernetes Engine (/kubernetes-engine) (GKE) and design a connected car platform on Google Kubernetes Engine (/kubernetes-engine).

"Our on-premises ERP system was purpose-built for our business needs so the ability to run our existing ERP licenses on Compute Engine is a game changer. We can scale services up and down much easier on the cloud than on-premises. Without hardware issues, staff can move from routine maintenance to focus on digital transformation."

—Sai Satheesh, Chief Technology Officer, TVS

Scaling resource-intensive applications on Compute Engine

ERP is critical for day-to-day business activities at TVS, from checking auto parts inventory to generating financial reports. Before Google Cloud, TVS managed more than 18 physical servers for their operating system, database, and backups, to support its legacy ERP system. This often led to hardware-related issues such as slow performance and system outage because the application ran out of disk space.

Additionally, TVS needed a nine-hour maintenance window, from 12 AM to 9 AM, to apply software updates and security patches for the numerous servers and databases. Backups had to be manually downloaded, with all files and data for the day saved on tape. Because of codependencies, applications connected to ERP, such as TVS PartSmart, its auto components business, are offline during the maintenance window to avoid losing new data.

By hosting its ERP on Compute Engine (/compute) virtual machines (VMs), TVS can simplify maintenance tasks and gain cloud scalability. For example, TVS can now backup data on Cloud Storage (/storage) in seconds instead of using a manual tape backup. The result? A six-hour maintenance window instead of nine. This 33% reduction in time means mechanics can order spare parts for customers' vehicles as early as 6 AM.

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To this end, TVS uses Cloud Logging (/logging) to log events and Cloud Monitoring (/monitoring) to monitor server performance to keep the network healthy. This helps the team quickly identify and resolve potential issues before they cause any business disruption.

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—Sai Satheesh, Chief Technology Officer, TVS

Uncovering data insights to drive business with Cloud Bigtable

TVS' success lies in its multi-brand car service, allowing TVS garages to service any brand or model. With an average of more than 30,000 parts per car, how do TVS garage owners and operators know what parts to stock to service every vehicle that arrives at their garage quickly?

The answer comes from data.

Using Cloud Bigtable (/bigtable), TVS analyzes more than 20 TB of transactional data such as service history, and real-time data such as warehouse inventory, to determine what parts are in stock or what to order based on seasonal patterns and user behavior. By forecasting demand, garages can reduce wait time for parts and get customers back on the road.

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Previously, TVS kept its sales catalog on a spreadsheet that was not user friendly and required manual updates. Niveus helped TVS create a sales catalog with Elasticsearch on GKE for its sales platform, TVS PartSmart. Suppliers can list their inventory through the app, where garage owners can then search and buy the auto parts by make, model, serial number, and other variables.

“Niveus’ team of Google-certified architects demonstrated their knowledge of Google technologies by delivering the sales catalog and the connected car platform. Despite very few in-person meetings, we collaborated well as a team,” says Satheesh.

"Data is the new oil in a digital economy. With data insights, we can improve customer interactions and generate revenue from our parts and service business. We need the power of Google Cloud for AI and analytics as we launch our connected car platform and onboard newer garages to our network in India and overseas."

—Sai Satheesh, Chief Technology Officer, TVS

Optimizing service delivery with connected car platform on Cloud IoT Core

TVS is now in the third stage of its digital transformation, as it further develops its connected car platform. It expects data consumption to jump by 10 TB to 15 TB, given the number of connected cars on the road and the data it plans to collect from each car.

By creating a connected ecosystem of car owners, garages, and parts suppliers, TVS aims to alert drivers about potential issues, so they know what to expect when they go for service. At the same time, the garages will know what parts they need before the car arrives for service. All TVS garages will be interconnected on the same garage management system (GMS), allowing mechanics to easily track maintenance history and update the customer interaction.

The connected car platform runs on IoT Core (/iot-core) in combination with other services on Google Cloud. A connected car dongle collects vehicle data such as speed and revolutions per minute via the on-board diagnostics (OBD) port. It streams the data pipeline via Dataflow (/dataflow) into Cloud Bigtable, which is ideal for analyzing large amounts of data. TVS intends to collect 50 to 60 data points per minute, per car. When a vehicle breaks down, the closest TVS garage knows whether to send roadside assistance for a quick repair or a tow truck to retrieve the car for extensive repairs.

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